Section 2.5-6

Linear Independence

Linear combination: Consider a vector space V and $x_1, x_2, \dots, x_k \in V$. Then every

$$v = \lambda_1 x_1 + \dots + \lambda_k x_k \quad , \text{ with } \lambda_1, \dots, \lambda_k \in \mathbb{R}$$

is a linear combination of $x_1, \dots, x_k$.

Ex: $x_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$ then $\begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix} = x_1 + 2x_2$ linear combination

Linear dependence / independence

For $x_1, \dots, x_k \in V$

(a) if there exists $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ s.t. at least one $\lambda_i \neq 0$ and $\sum_{i=1}^{k} \lambda_i x_i = 0$ then the vectors $x_1, \dots, x_k$ are linearly dependent.

(b) if the solution to $\sum_{i=1}^{k} \lambda_i x_i = 0$ is only the trivial solution $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$ then $x_1, \dots, x_k$ are linearly independent.

Ex 1: $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$ are linearly dependent because $\begin{bmatrix}1\\2\\1\end{bmatrix} + 2\begin{bmatrix}0\\1\\-1\end{bmatrix} - \begin{bmatrix}1\\4\\-1\end{bmatrix} = 0$

Ex 2: $\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}0\\1\\-1\end{bmatrix}$ are linearly independent

Note that:

- $x_1, \dots, x_k$ are linearly dependent if and only if one of them is a linear combination of the other vectors.
- if $x_i = c x_j$ then they must be linearly dependent.
- Apply Gaussian elimination to $\begin{bmatrix} x_1 & x_2 & \cdots & x_k \end{bmatrix}$

until it is row echelon form.

a) the pivot columns indicate the vectors that are linearly independent.

b) non-pivot collumns can be written as a linear combination of pivot columns.

$$\begin{bmatrix} x & x & x & x & x \\ 0 & x & x & x & x \\ 0 & 0 & 0 & x & x \\ 0 & 0 & 0 & 0 & x \end{bmatrix} \qquad \begin{bmatrix} 0 & x & x & x & x & x \\ 0 & 0 & 0 & x & x & x \\ 0 & 0 & 0 & x & x & x \\ 0 & 0 & 0 & 0 & 0 & x \end{bmatrix}$$

Ex1:

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Ex2:

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix} \qquad x_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix} \qquad x_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & -2 \\ -3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix} \xrightarrow[R_4 \leftarrow R_4 - 4R_1]{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 + 3R_1}} \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 0 \\ 0 & 3 & -2 \\ 0 & -2 & 5 \end{bmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 + 3R_2}$$

pivots

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

Remark: a) m linear combination of k vectors are lin. dependent if $m > k$.

b) m vectors in $\mathbb{R}^k$ are lin. dependent if $m > k$.

Basis and Rank (2.6)

Definition of generating set and span:

Consider a vector space V and a set $A = \{x_1, \dots, x_k\} \subseteq V$.

(a) If every $v \in V$ is a linear combination of $x_1, \dots, x_k$ then A is a generating set of V.

(b) The set of all linear combination of $x_1, \dots, x_k$ is called span of A, denoted by $\mathrm{Span}(A)$ or $\mathrm{Span}(x_1, \dots, x_k)$.

Remark: If $\mathrm{Span}(A) = V$, then A is a generating set of V.

Ex 1: $V = \mathbb{R}^2$ a) $A = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is a generating set of V

b) $A = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is a generating set of V

c) $A = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is not a generating set of V

Ex 2: V = set of all polynomials

$A = \{1, x, x^2, \dots, x^n, \dots\}$ is a generating set of V.

Definition of Basis:

A generating set A is called minimal if there does not exist a smaller set $\tilde{A}$ s.t. $\tilde{A} \subset A$ and $\tilde{A}$ is another generating set. Every minimal generating set of V is called a basis.

Remark:

- Every linearly independent generating set is minimal and a basis.
- If B is a maximal linearly independent set of vectors in V, i.e., adding any vector to B will make it linearly dependent, then B is a basis.
- Every vector $x \in V$ is a linear combination of the vectors in B, and the combination is unique.

Examples 3

a) In $\mathbb{R}^3$, the standard basis is $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix} \Rightarrow \underset{3}{\underset{\parallel}{\operatorname{rank}(\mathbb{R}^3)}}$

b) other basis: $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$.

c) $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$ are linearly independent, but not a basis.

Back to Ex **1**:

a) $A = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}$ not a basis

b) $A = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\} \rightarrow$ standard basis for $\mathbb{R}^2$

c) $A = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ not a basis

Ex 2 (book)

$\{1, x, x^2, \ldots\}$ basis for the space of poly.

In general, the basis for the subspace $\text{Span}\{x_1, \ldots, x_m\} \subseteq \mathbb{R}^n$ can be found by doing:

- find the row echelon form of $[x_1 \cdots x_m] \in \mathbb{R}^{n \times m}$
- find the vectors associated with the pivot columns of the echelon form.

Rank (Definition)

For a matrix $A \in \mathbb{R}^{m \times n}$, the rank is the number of linearly independent columns (rows). We write rank(A).

Ex 4:

$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ rank (A) = 2 $\Rightarrow$ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

col 3 = col 1 + col 2

row 3 = 0 row 1.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix} \xrightarrow[R3 \leftarrow R3 - 3R1]{R2 \leftarrow R2 + 2R1} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & -1 & -3 \end{bmatrix} \xrightarrow[R3 \leftarrow R3 + R2]{} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

rank$(A) = 2$ → two pivots

Properties

- rank$(A) =$ rank(A^T)
- columns of $A \in \mathbb{R}^{m \times n}$ spans a subspace in $\mathbb{R}^m$,
  rows $\mathbb{R}^n$,

dimension rank(A).

- $A \in \mathbb{R}^{n \times n}$ is invertible if and only if rank$(A) = n$
- For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $Ax = b$ can be solved if and only if rank$(A) =$ rank$([A\ b])$

augmented matrix $\in \mathbb{R}^{m \times (n+1)}$

- $\{x \mid Ax = 0\}$ is a subspace with dimension $n -$ rank(A)

(dimension of V: the number of vectors in a basis of V).

- A is called full rank if rank$(A) =$ largest possible rank for $m \times n$ matrices $= \min(m, n)$.

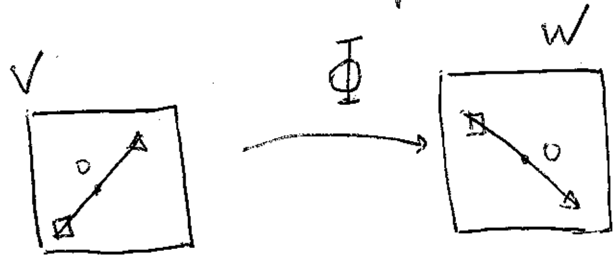
Section 2.7.

Linear Mapping

Consider two vector spaces V, W. A mapping $\Phi: V \to W$ is a linear mapping if $\forall x, y \in V$, $\forall \lambda, \phi \in \mathbb{R}$, we have

$$\Phi(\lambda x + \phi y) = \lambda \Phi(x) + \phi \Phi(y)$$

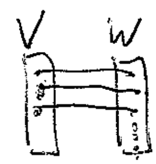
Note that it preserves structure, it maps a subspace straightline to another subspace straightline.



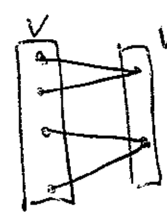
Definition (Injective, Surjective, Bijective)

Consider $\Phi: V \to W$, where V, W can be arbitrary sets. Then Φ is called

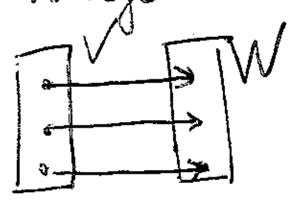
a) Injective: if $\forall x, y \in V$ $\Phi(x) = \Phi(y) \Rightarrow x = y$



b) Surjective: if $\Phi(V) = W$



c) Bijective: if both injective and surjective



Bijective + linear = isomorphism

Bijective + linear + $(\Phi: V \to V)$ = Automorphism

linear + $(\Phi: V \to V)$ = endomorphism

$id_V : V \to V$ $\quad id_V(x) = x \to$ identity mapping

- For linear mappings $\Phi: V \to W$ and $\Psi: W \to X$ Then $\Psi \circ \Phi$ is linear.

- If $\Phi: V \to W$ is an isomorphism, then $\Phi^{-1}: W \to V$ exists and it is also an isomorphism.

<u>Thm</u>] Finite-dimensional vector spaces V and W are isomorphic if and only if $\dim(V) = \dim(W)$.

$\uparrow$ dimension.

This means that vector spaces of the same dimension are kind of the same thing.

<u>Matrix Representation of linear mapping</u>

Assume V is n-dimensional, a ordered basis is $B = (b_1, \dots, b_n)$

Ex 1: $V = \mathbb{R}^n \to$ $e_1 = (1, 0, 0, 0, \dots 0)$

$e_2 = (0, 1, 0, \dots 0)$

$e_n = (0, 0, \dots, 0, 1)$

<u>Definition</u> (Coordinates):

For any $x \in V$, there exists a unique representation

$$x = \alpha_1 b_1 + \dots + \alpha_n b_n$$

Then $(\alpha_1, \dots, \alpha_n)$ are the coordinates of x with respect to B, and $\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$ is the coordinate vector.

Ex 2: $V = \mathbb{R}^2$

$B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ $\quad x = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \longrightarrow$ coord. vector $= \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ $\longrightarrow$ coord vector $= \begin{bmatrix} 5/2 \\ -1/2 \end{bmatrix}$

<u>Def</u> (Transformation Matrix)

Consider vector spaces V, W with corresponding ordered bases $B = (b_1, \dots, b_n)$ and $C = (c_1, \dots, c_m)$

Consider a linear mapping $\Phi : V \to W$. For $j = 1, \dots, n$,

$$\Phi(b_j) = \alpha_{1j} c_1 + \dots + \alpha_{mj} c_m = \sum_{i=1}^{m} \alpha_{ij} c_i$$

is the unique representation of $\Phi(b_j)$ with respect to C. The matrix $A_\Phi \in \mathbb{R}^{m \times n}$ with entries

$$A_\Phi(i,j) = \alpha_{ij}$$

is called the transformation matrix of Φ with respect to the bases B of V and C of W.

- If $\hat{x}$ is the coordinate vector of $x \in V$, and $\hat{y}$ is the coordinate vector of $\Phi(x) \in W$, then

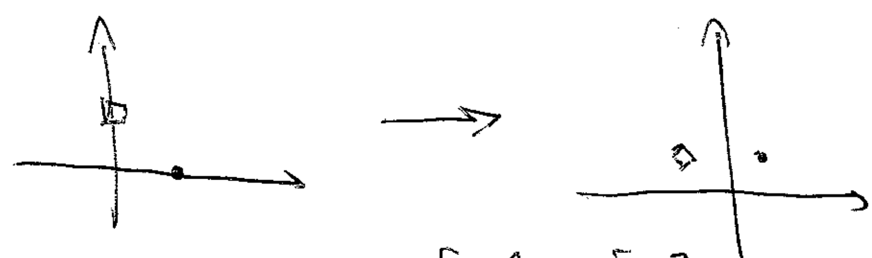
$$\hat{y} = A_\Phi(\hat{x})$$

Example: $V, W = \mathbb{R}^2$ $\quad B = C = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

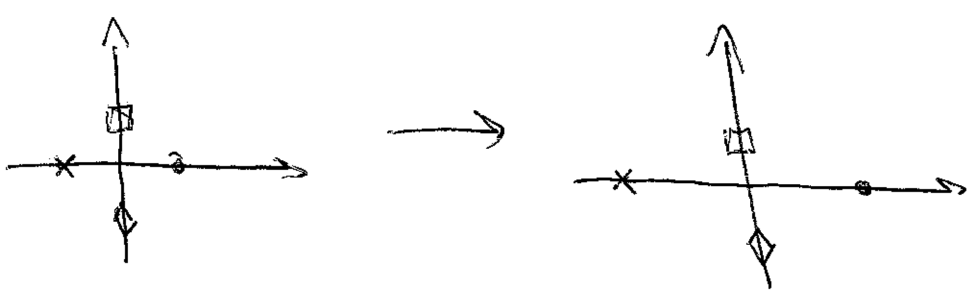
$$A_\Phi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \implies y = A_\Phi x = x$$

$$A_\Phi = \begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{bmatrix} = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

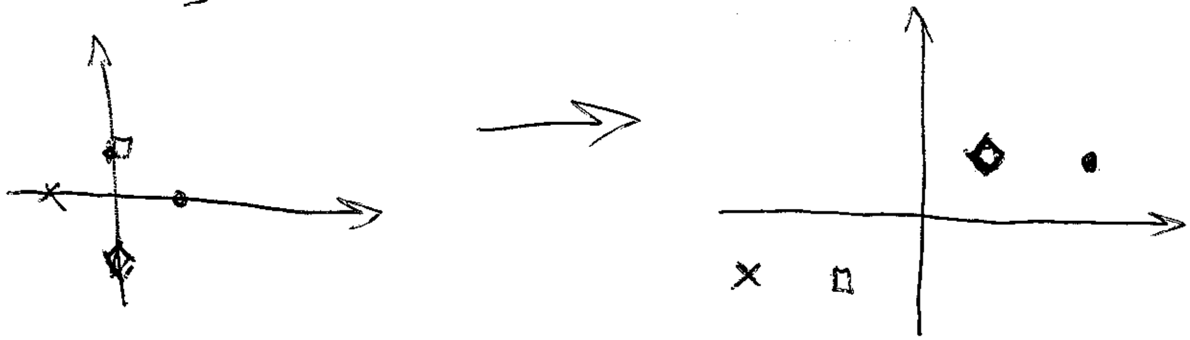
$$A_\Phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} \qquad A_\Phi \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$



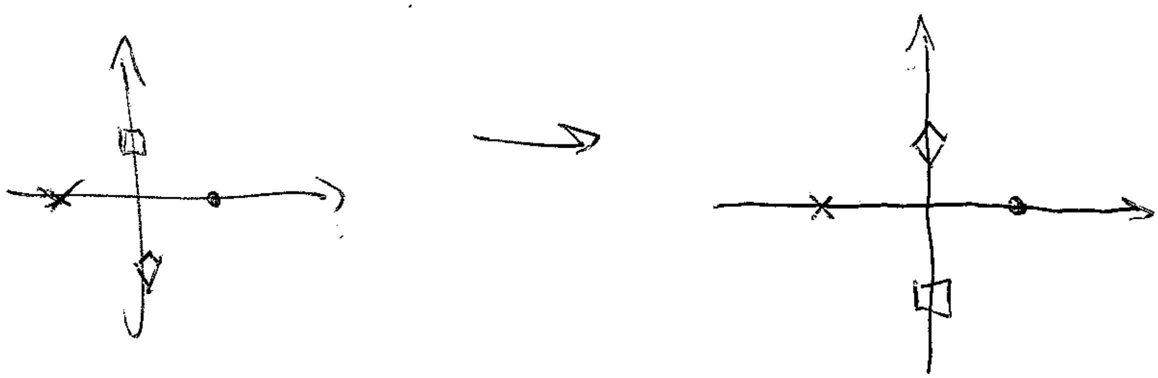
$$A_\Phi = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad A_\Phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad A_\Phi \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$A_\Phi = \begin{bmatrix} 3 & -1 \\ 1 & -1 \end{bmatrix} \qquad A_\Phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \qquad A_\Phi \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$



$$A_\Phi = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \qquad A_\Phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad A_\Phi \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$



Ex: Take $\Phi : V \to W$ $\quad B = (b_1, b_2, b_3)$ basis for V

$C = (c_1, c_2, c_3, c_4)$ basis for W

$$\Phi(b_1) = c_1 - c_2 + 3c_3 - c_4$$
$$\Phi(b_2) = 2c_1 + c_2 + 7c_3 + 2c_4$$
$$\Phi(b_3) = 3c_2 + c_3 + 4c_4$$

Who is A_Φ, the transformation matrix with respect to B and C?

$$A_\Phi = [\alpha_1\ \alpha_2\ \alpha_3] = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \\ 3 & 7 & 1 \\ -1 & 2 & 4 \end{bmatrix}$$

$$\alpha_j = \Phi(b_j) \rightarrow \Phi(b_j) = \sum_{i=1}^{4} \alpha_{ij} b_j$$

Image and Kernel

Def For $\Phi : V \rightarrow W$, we define the Kernel/nullspace

$$\text{Ker}\{\Phi\} := \Phi^{-1}(0_W) = \{v \in V : \Phi(v) = 0_W\}$$

and the image/range

$$\text{Im}\{\Phi\} := \Phi(V) = \{w \in W \mid \exists v \in V : \Phi(v) = w\}$$

We also call V and W the domain and codomain of Φ, respectively.

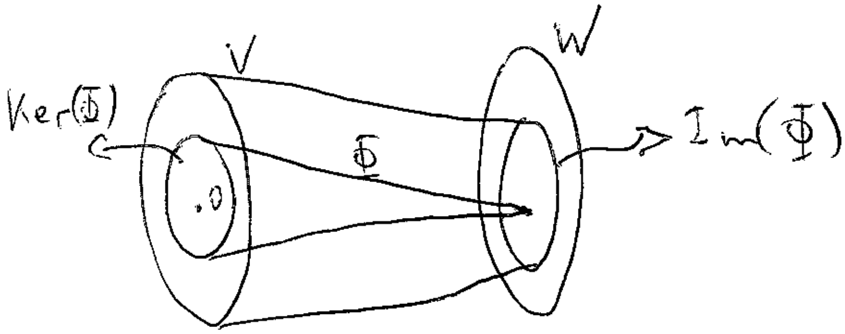
- Since $\Phi(0) = 0 \Rightarrow 0 \in \text{Ker}(\Phi)$

$$\Downarrow$$

$$\Phi(0) = \Phi(0 \cdot v) = 0 \cdot \Phi(v) = 0$$

- $\text{Im}(\Phi)$ is a subspace of W

  $\text{Ker}(\Phi)$ is a subspace of V



- Φ is injective if and only if $\operatorname{Ker}(\Phi) = 0$
- $\operatorname{rank}(A) = \dim(\operatorname{Im}(\Phi))$
- Kernel can be used to determine how we express a column of A as a linear combination of other columns.

Example

$\Phi : \mathbb{R}^4 \to \mathbb{R}^3 \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$

$$\Phi(x) = \begin{bmatrix} x_1 + x_2 + 2x_3 + 3x_4 \\ 3x_1 + 4x_2 - x_3 + 2x_4 \\ -x_1 - 2x_2 + 5x_3 + 4x_4 \end{bmatrix}$$

$$= A x = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 4 & -1 & 2 \\ -1 & -2 & 5 & 4 \end{bmatrix} x$$

$$\operatorname{Im}(\Phi) = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \right\}$$

(basis: vectors associated with pivot columns in echelon form)

$$A \xrightarrow[R_3 = R_3 + R_1]{R_2 = R_2 - 3R_1} \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & -7 & -7 \\ 0 & -1 & 7 & 7 \end{bmatrix} \xrightarrow{R_3 = R_3 - R_2} \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & -7 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} \right\}$$

basis

$$\operatorname{Ker}(\Phi) = \{ v \mid \Phi(v) = 0 \} = \{ v \in \mathbb{R}^4 \mid Av = 0 \}$$

free variables

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & -7 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

2nd eq

$$v_2 = 7v_3 + 7v_4$$

1st eq

$$v_1 + v_2 + 2v_3 + 3v_4 = 0$$

$$v_1 = -v_2 - 2v_3 - 3v_4$$

$$= -9v_3 - 10v_4$$

Then $$v = \begin{bmatrix} -9v_3 - 10v_4 \\ 7v_3 + 7v_4 \\ v_3 \\ v_4 \end{bmatrix} = v_3 \begin{bmatrix} -9 \\ 7 \\ 1 \\ 0 \end{bmatrix} + v_4 \begin{bmatrix} -10 \\ 7 \\ 0 \\ 1 \end{bmatrix}$$

$v_3, v_4 \in \mathbb{R}$.

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theorem) (Rank-Nullity Theorem)

For vector spaces V, W and a linear mapping $\Phi: V \to W$ it holds that

$$\dim\{Ker(\Phi)\} + \dim\{Im(\Phi)\} = \dim(V)$$

Note that for the previous example

$\dim V = 4 \to V = \mathbb{R}^4$

$\dim\{Ker(\Phi)\} = 2 \quad \dim\{Im(\Phi)\} = 2$

Remarks:

1) If $\dim(Im(\Phi)) < \dim(V)$, then the Kernel is non-trivial ($\dim(Ker(\Phi)) \geq 1$).

2) If $\dim(Im(\Phi)) < \dim(V)$ and A_Φ is the transf. matrix, then $A_\Phi x = 0$ has inf. many solutions

3) If $\dim(V) = \dim(W)$, then the following are equivalent

a) Φ is injective

b) Φ is surjective

c) Φ is bijective

Proof: (a) $\to$ (b)

If $v \neq 0$, $v \in Ker\{0\}$ then $\Phi(v) = 0$

$\Phi(x) = \Phi(x) + \Phi(v) = \Phi(x+v)$

but $x \neq x+v$

contradicts injectivity.

Then $Ker(\Phi) = \{0\} \to \dim(Ker(\Phi)) = 0$

$\dim(Im(\Phi)) = \dim V = \dim W \Rightarrow Im(\Phi) = W$.

Φ is surjective

Change of Basis

Thm) If $\Phi : V \longrightarrow W$ has a transformation matrix A_Φ with respect to the basis B in V and basis C in W and $\tilde{A}_\Phi$ with respect to the basis $\tilde{B}$ in V and basis $\tilde{C}$ in W, then

$$\tilde{A}_\Phi = T^{-1} A_\Phi S$$

where T is the transformation of the identity mapping from coordinates w.r.t. $\tilde{B}$ to coordinates w.r.t. B and S is the transformation matrix of the identity mapping from coordinates w.r.t. $\tilde{C}$ to coordinates w.r.t. C.

Ex: Consider $\Phi : \mathbb{R}^3 \to \mathbb{R}^4$ a linear mapping with transf. matrix

$$A_\Phi = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \\ 3 & 7 & 1 \\ -1 & 2 & 4 \end{bmatrix}$$

with respect to the Basis

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ and } C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Find $\tilde{A}_\Phi$ of Φ w.r.t. to the Bases

$$\tilde{\mathcal{B}} = \left\{ \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix} \right\}, \quad \mathcal{C} = \left\{ \begin{bmatrix}1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\\1\end{bmatrix} \right\}$$

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We have that

$$S = \begin{bmatrix}1 & 0 & 1\\1 & 1 & 0\\0 & 1 & 1\end{bmatrix} \qquad T = \begin{bmatrix}1 & 1 & 0 & 1\\1 & 0 & 1 & 0\\0 & 1 & 1 & 0\\0 & 0 & 0 & 1\end{bmatrix}$$

i^{th} column of S is the coordinate representation of $\tilde{b}_i$ in terms of $\mathcal{B}$.

i^{th} column of T is the coordinate representation of $\tilde{c}_i$ in terms of C.

$$\tilde{A}_\Phi = T^{-1} A_\Phi S = \begin{bmatrix}1/2 & 1/2 & -1/2 & -1/2\\1/2 & -1/2 & 1/2 & -1/2\\-1/2 & 1/2 & 1/2 & 1/2\\0 & 0 & 0 & 1\end{bmatrix} \begin{bmatrix}1 & 2 & 0\\-1 & 1 & 3\\3 & 7 & 1\\1 & 2 & 4\end{bmatrix} \begin{bmatrix}1 & 0 & 1\\1 & 1 & 0\\0 & 1 & 1\end{bmatrix}$$

$$= \begin{bmatrix}-4 & -4 & -2\\6 & 0 & 0\\4 & 8 & 4\\1 & 6 & 3\end{bmatrix}$$